

Lincoln University College
Faculty of Computer Science and Multimedia

Cafe Management System
A Project Proposal

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Department of Information Technology
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Contents

1. Introduction
2. Problem Statement
3. Objectives
4. Methodology
 - 4.1 Requirement Identification
 - 4.1.1 Study of Existing System
 - 4.1.2 Literature Review
 - 4.1.3 Requirement Analysis
 - 4.2 Feasibility Study
 - 4.2.1 Technical Feasibility
 - 4.2.2 Operational Feasibility
 - 4.2.3 Economic Feasibility
 - 4.3 High Level Design of System
 - 4.4 Flowchart
 - 4.5 Gantt Chart
5. Expected Outcome
6. References

1. Introduction

The cafe industry has grown rapidly in recent years, especially in urban areas of Nepal, with increasing demand for fast, efficient, and quality service. Most small and medium-sized cafes still rely on manual processes such as handwritten order taking, manual billing, and informal inventory tracking. These practices often lead to order delays, billing errors, poor inventory control, and difficulty in managing customer data.

A **Cafe Management System (CMS)** is a digital solution designed to automate daily cafe operations including order management, billing, menu handling, inventory tracking, and report generation. The proposed system aims to provide a simple, user-friendly platform for café owners, staff, and administrators to manage operations efficiently and improve customer satisfaction.

2. Problem Statement

Many cafes currently use manual or semi-digital systems to manage orders and billing, which results in inefficiencies such as incorrect orders, slow service, lack of real-time sales tracking, and difficulty in managing inventory. During peak

hours, these issues become more severe, affecting customer experience and business profitability.

Additionally, cafe owners lack proper analytical tools to monitor sales performance, popular menu items, and daily revenue. The absence of an integrated system makes decision-making difficult and increases operational costs. Therefore, there is a strong need for a centralized, automated Cafe Management System that streamlines operations, reduces human error, and improves overall efficiency.

3. Objectives

The objectives of this project are:

- To develop a digital system for managing café orders and billing
- To automate menu management and inventory tracking
- To reduce manual errors in order processing and payment
- To provide real-time sales and performance reports
- To improve service speed and customer satisfaction

4. Methodology

4.1 Requirement Identification

4.1.1 Study of Existing System

Currently, most cafes use manual order-taking methods where waiters write orders on paper and submit them to the

kitchen. Billing is often done manually or using basic calculators. Inventory tracking is either not maintained or recorded separately in notebooks.

Some cafes use POS systems, but these are often expensive, complex, or not customized for local cafe needs. Small cafes cannot afford such systems, leading to inefficient operations. These limitations highlight the need for a cost-effective, simple, and customizable cafe management solution.

4.1.2 Literature Review

Various studies on restaurant and cafe management systems emphasize the importance of automation in improving service quality and operational efficiency. Research papers such as “*Restaurant Management System using Web Technology*” (IJCSIT, 2023) highlight how digital ordering and billing systems reduce errors and improve customer satisfaction.

Other studies suggest that real-time inventory and sales reporting help business owners make informed decisions. Existing systems in developed markets show high efficiency, but they are often not adapted to local business practices and budget constraints. This project focuses on developing a system suitable for local cafés in Nepal.

4.1.3 Requirement Analysis

Functional Requirements

1. User login and authentication
2. Menu management (add, update, delete items)
3. Order placement and order tracking
4. Billing and invoice generation
5. Inventory management
6. Sales and daily report generation
7. Admin dashboard
8. Staff management

Non-Functional Requirements

1. Usability: Simple and intuitive interface
2. Performance: Fast order processing
3. Reliability: Accurate data handling
4. Security: Role-based access control
5. Scalability: Can support multiple café branches
6. Availability: System should be accessible during working hours

4.2 Feasibility Study

4.2.1 Technical Feasibility

The system can be developed using widely available technologies such as HTML, CSS, JavaScript, React or Flutter for frontend, and Node.js with Firebase/MySQL for backend. These technologies support CRUD operations and real-time updates, making the project technically feasible.

4.2.2 Operational Feasibility

Cafe staff can easily adapt to the system due to its simple UI. Minimal training is required to operate the system. Since similar apps are already used in daily life, user acceptance is expected to be high.

4.2.3 Economic Feasibility

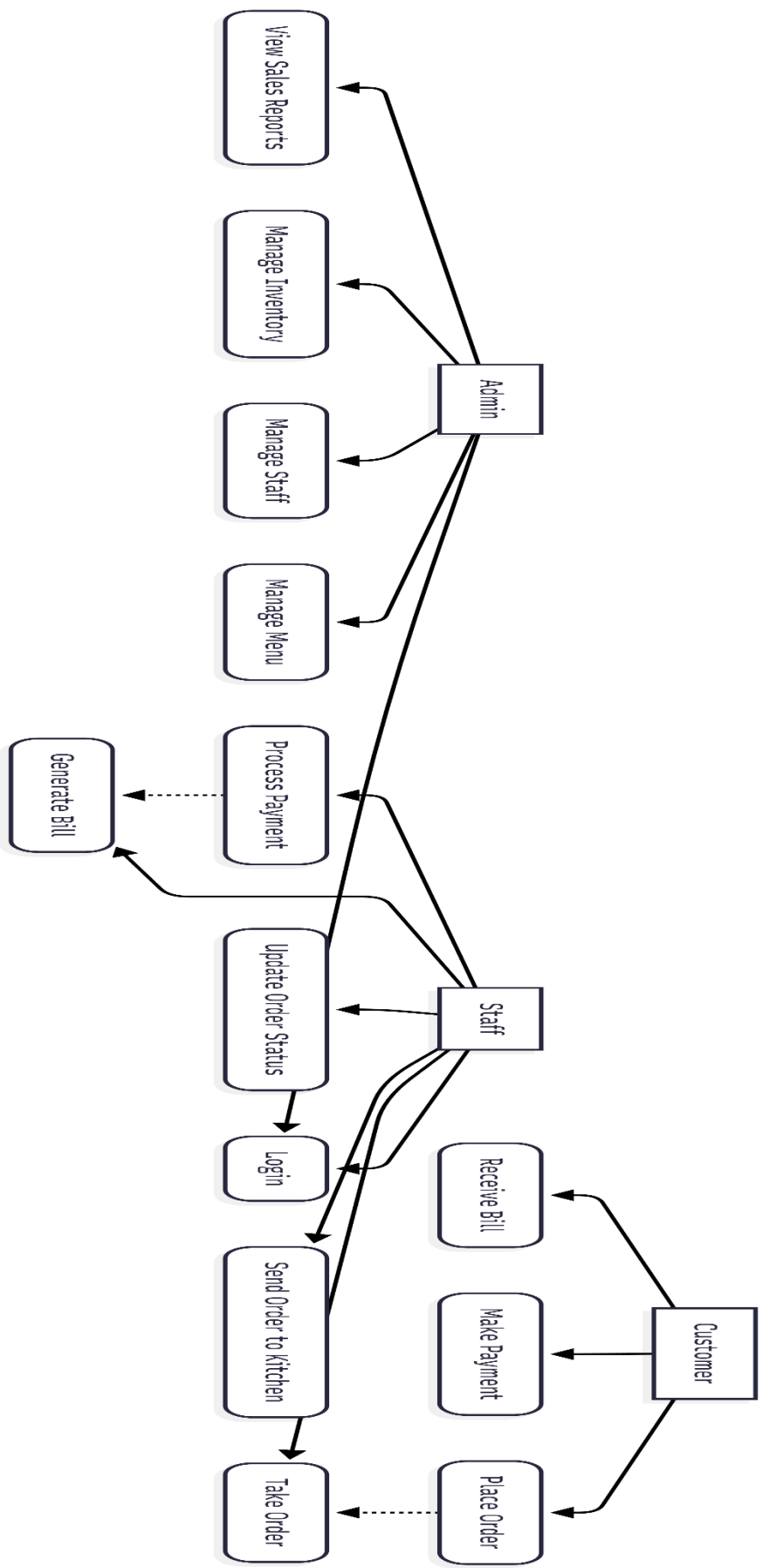
The system uses open-source tools and frameworks, reducing development cost. No special hardware is required except basic computers or tablets. Hosting can be done on low-cost servers, making the project economically feasible.

4.3 High Level Design of System

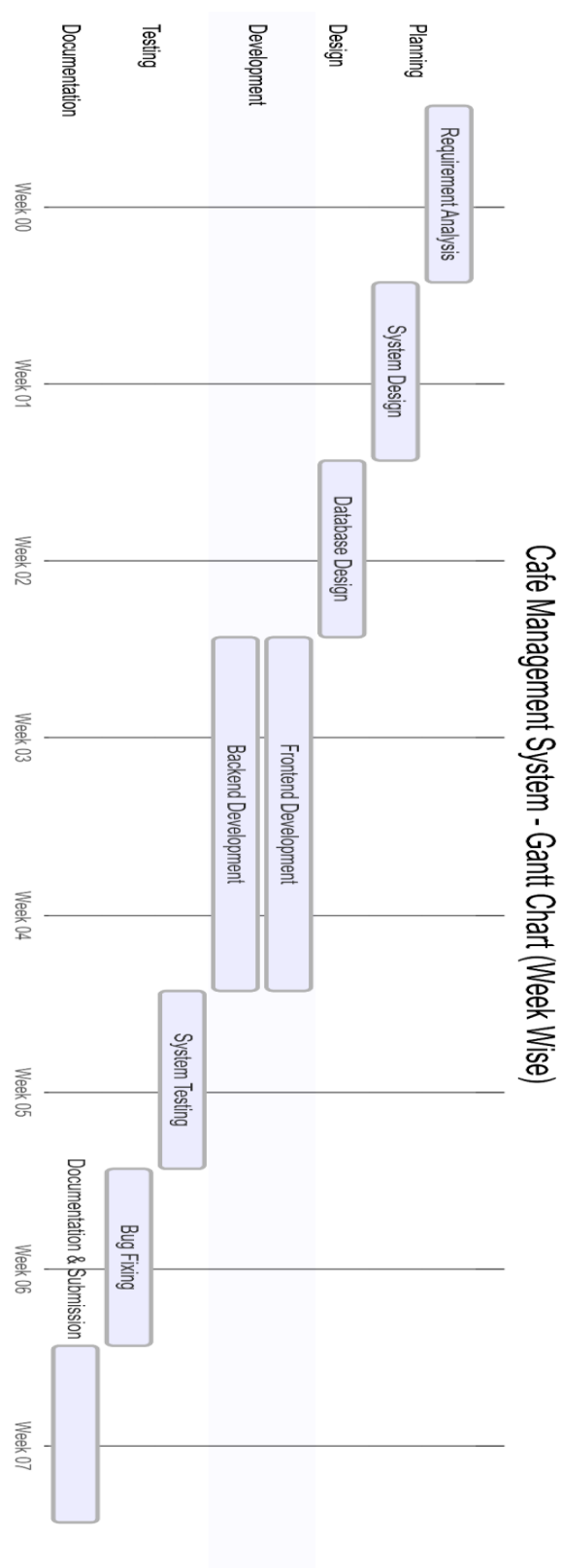
The system consists of three main modules:

- **Admin Module** – manages menu, users, reports
- **Staff Module** – handles order placement and billing
- **System Database** – stores orders, inventory, and reports

4.4 Flowchart (Cafe Management System)



4.5 Gantt Chart



5. Expected Outcome

The expected outcome of this project is a fully functional Cafe Management System that automates ordering, billing, and inventory management. The system will improve service speed, reduce manual errors, and provide real-time business insights. It will be suitable for small and medium cafés and demonstrate a practical real-world application of information technology.

6. References

1. IJCSIT (2023). Restaurant Management System using Web Technology
2. IJRASET (2022). Café Billing and Order Management System
3. IEEE (2021). POS and Inventory Management Systems
4. Online Articles on Restaurant Automation Systems